

Computational Modeling of Tandem Plunging Airfoils: A Lumped Vortex Method for Unsteady Aerodynamics

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Abstract

This report presents the development and implementation of a computational framework for modeling the unsteady aerodynamics of tandem plunging airfoils using the Lumped Vortex Element Method (LVEM). The method treats each airfoil as a single bound vortex at the quarter-chord with a collocation point at three-quarter chord, coupled with a Lagrangian wake convection model. The framework captures complex aerodynamic interactions including wake-body interference, vortex-induced velocities, and the transition between drag-producing and thrust-producing regimes as a function of the reduced frequency-amplitude product (kh). Validation against classical unsteady airfoil theory and parametric studies demonstrate the method's capability to predict force coefficients and wake dynamics for bio-inspired propulsion systems and formation flight configurations.

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Chapter 1

Introduction

1.1 Motivation and Background

Unsteady aerodynamics governs the flight of biological flyers such as birds, bats, and insects, as well as modern Micro Air Vehicles (MAVs) and flapping-wing systems. Unlike steady-state aerodynamics where forces can be predicted using quasi-steady assumptions, unsteady flows involve rapid changes in angle of attack, time-varying circulation, and complex wake structures that fundamentally alter the aerodynamic loading.

One of the most intriguing phenomena in unsteady aerodynamics is the ability of oscillating airfoils to generate thrust rather than drag. This thrust production depends critically on the **Strouhal number** (St) or equivalently the product of reduced frequency and amplitude (kh). When kh exceeds approximately 0.2-0.4, the wake structure transitions from a drag-producing von Kármán vortex street (staggered vortices) to a thrust-producing reverse Kármán street (aligned vortices).

1.2 Tandem Configuration and Wake Surfing

In nature and engineering applications, multiple lifting surfaces often operate in close proximity. Formation flight of birds demonstrates energy-saving benefits from favorable wake interactions. Similarly, tandem wing configurations in aircraft and the interaction between forewing and hindwing in dragonflies exploit wake interference for enhanced performance.

The downstream airfoil in a tandem configuration experiences:

- Modified effective freestream velocity due to upstream wake
- Periodic gusts from shed vortices
- Altered effective angle of attack
- Potential for constructive or destructive interference depending on phase relationship

This phenomenon, sometimes called "wake surfing" or "vortex surfing," can significantly enhance or degrade performance depending on the relative motion parameters.

1.3 Objectives

The primary objectives of this project are:

1. Develop a computationally efficient Lumped Vortex Element Method for tandem plunging airfoils
2. Validate the solver against analytical theory for single airfoils
3. Investigate the effect of the kh parameter on thrust/drag production
4. Analyze wake interference effects in tandem configurations
5. Demonstrate the importance of numerical regularization techniques (Lamb-Oseen cores)

Chapter 2

Theoretical Foundation

2.1 Theoretical Foundation

2.1.1 Potential Flow Theory

The computational framework is built upon the assumption that the flow field is incompressible ($\nabla \cdot \vec{V} = 0$) and irrotational ($\nabla \times \vec{V} = 0$) everywhere except at discrete singular points containing vorticity. Under these assumptions, the velocity field $\vec{V}$ can be described by a scalar velocity potential $\Phi(x, y, t)$ such that the velocity vector is the gradient of the potential:

$$\vec{V} = \nabla \Phi \quad (2.1.1)$$

Substituting this velocity potential into the incompressible continuity equation yields Laplace's equation, which serves as the governing linear differential equation for the entire fluid domain:

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0 \quad (2.1.2)$$

Because Laplace's equation is a linear partial differential equation, the principle of superposition applies. This is the fundamental mathematical justification for the Discrete Vortex Method: complex flow fields around tandem airfoils can be constructed simply by linearly superimposing elementary solutions, specifically a uniform freestream and discrete point vortices.

The Point Vortex and Biot-Savart Law Derivation

In two-dimensional flow, a point vortex is a fundamental solution to Laplace's equation where all vorticity ($\vec{\omega}$) is concentrated at a single singularity. The strength of this vortex is defined by its circulation, Γ , which is the macroscopic line integral of the velocity field along any closed contour C enclosing the core:

$$\Gamma = \oint_C \vec{V} \cdot d\vec{l} = \iint_A \vec{\omega} \cdot d\vec{A} \quad (2.1.3)$$

Assuming a circular fluid contour of radius r centered exactly on the vortex, the induced flow is purely tangential ($V_r = 0$, $V = V_\theta$). The line integral simplifies algebraically

to the circumference of the circle multiplied by the tangential velocity:

$$\Gamma = \int_0^{2\pi} V_\theta(r) d\theta = V_\theta(2\pi r) \implies V_\theta = \frac{\Gamma}{2\pi r} \quad (2.1.4)$$

To implement this computationally in the solver, this scalar tangential velocity must be converted into a Cartesian vector field. For a vortex located at an arbitrary point (x_v, y_v) inducing a velocity at a target point (x_p, y_p) , we define the distance vector $\vec{r} = (x_p - x_v)\hat{i} + (y_p - y_v)\hat{j}$. Applying the right-hand rule via a cross product yields the classical 2D Biot-Savart Law:

$$\vec{V}_{induced} = \frac{\Gamma}{2\pi} \frac{\hat{k} \times \vec{r}}{|\vec{r}|^2} \quad (2.1.5)$$

This equation is the primary engine of the computational framework, continuously calculating the velocity induced by every bound panel vortex and every free wake vortex throughout the simulation.

2.1.2 The Point Vortex and Biot-Savart Law Derivation

In two-dimensional flow, a point vortex is a fundamental solution to Laplace's equation where all vorticity ($\vec{\omega}$) is concentrated at a single singular point. The circulation Γ of this vortex is defined as the line integral of the velocity around any closed curve enclosing the core:

$$\Gamma = \oint_C \vec{V} \cdot d\vec{l} = \iint_A \vec{\omega} \cdot d\vec{A} \quad (2.1.6)$$

For a point vortex located at the origin, the flow is purely tangential ($V_r = 0$). Evaluating the line integral along a circle of radius r yields the induced tangential velocity:

$$\Gamma = V_\theta(2\pi r) \implies V_\theta = \frac{\Gamma}{2\pi r} \quad (2.1.7)$$

Transforming this into Cartesian coordinates for a vortex located at an arbitrary point (x_v, y_v) inducing a velocity at a target point (x_p, y_p) , we arrive at the standard 2D Biot-Savart Law. In cross-product vector notation, this is:

$$\vec{V}_{induced} = \frac{\Gamma}{2\pi} \frac{\hat{k} \times \vec{r}}{|\vec{r}|^2} \quad (2.1.8)$$

where $\vec{r} = (x_p - x_v)\hat{i} + (y_p - y_v)\hat{j}$ is the distance vector.

2.1.3 Implementation of Viscous Regularization (Lamb-Oseen Core)

A critical limitation of the pure potential point vortex (Equation 5) is the singularity at $r \rightarrow 0$, where the induced velocity approaches infinity. In a discrete numerical simulation,

this causes catastrophic instabilities if wake particles convect too close to one another.

To resolve this, the code implements a **Lamb-Oseen viscous core model**. The Lamb-Oseen vortex is an exact analytical solution to the incompressible Navier-Stokes equations for a decaying line vortex. It introduces viscous diffusion, distributing the vorticity over a finite core radius $r_c(t)$. The exact tangential velocity profile of a Lamb-Oseen vortex is:

$$V_\theta(r, t) = \frac{\Gamma}{2\pi r} \left(1 - \exp \left(-\frac{r^2}{r_c(t)^2} \right) \right) \quad (2.1.9)$$

where the core radius grows over time due to the fluid's kinematic viscosity (ν):

$$r_c(t) = \sqrt{4\nu t + r_{c,0}^2} \quad (2.1.10)$$

In the computational implementation, rather than computing the computationally expensive exponential function, a standard algebraic regularization parameter is added to the denominator of the Biot-Savart law. This achieves the same desingularization effect with superior computational efficiency:

$$\vec{V}_{induced} = \frac{\Gamma}{2\pi} \frac{\hat{k} \times \vec{r}}{|\vec{r}|^2 + r_c^2} \quad (2.1.11)$$

This regularization ensures $V \rightarrow 0$ as $r \rightarrow 0$, mirroring true fluid viscosity and stabilizing the tandem wake interactions.

2.1.4 Kelvin's Theorem and the Unsteady Kutta Condition

For an airfoil generating lift, there must be a net circulation around it. However, the fluid begins from rest with zero total circulation. According to Kelvin's Circulation Theorem, for an inviscid, barotropic fluid subject to conservative forces, the material derivative of circulation around a closed fluid contour is zero:

$$\frac{D\Gamma_{total}}{Dt} = 0 \implies \Gamma_{bound}(t) + \sum \Gamma_{wake}(t) = 0 \quad (2.1.12)$$

This theorem fundamentally mandates the existence of the wake. When the plunging airfoil rapidly changes its angle of attack, its bound circulation (Γ_{bound}) changes. To maintain $\Gamma_{total} = 0$, a counter-rotating "starting vortex" must be shed from the trailing edge.

In the Unsteady Discrete Vortex Method (UDVM) implementation, this is mathematically enforced at every time step (Δt) to satisfy the **Unsteady Kutta Condition**. The Kutta condition dictates that the flow must leave the sharp trailing edge smoothly without infinite velocities. By shedding exactly one vortex particle per time step, the

instantaneous jump in bound circulation is neutralized at the trailing edge:

$$\Gamma_{shed}^{(t)} = \sum_{j=1}^N \Gamma_{bound,j}^{(t-\Delta t)} - \sum_{j=1}^N \Gamma_{bound,j}^{(t)} \quad (2.1.13)$$

2.1.5 Formulation of the Boundary Value Problem

To determine the unknown bound circulations ($\Gamma_{bound,j}$) on the airfoil, the theoretical flow tangency boundary condition must be implemented. For a solid body moving through a fluid, the fluid cannot penetrate the surface. Mathematically, the relative normal velocity between the fluid and the body surface must be zero:

$$(\vec{V}_{fluid} - \vec{V}_{body}) \cdot \vec{n} = 0 \quad (2.1.14)$$

Expanding the fluid velocity into its components (freestream, wake-induced, and bound-vortex-induced), we arrive at the continuous boundary integral equation:

$$\left(\vec{U}_{\infty} + \iint_{wake} \vec{V}_{ind} d\Gamma_w + \oint_{airfoil} \vec{V}_{ind} d\Gamma_b - \vec{V}_{kinematic} \right) \cdot \vec{n} = 0 \quad (2.1.15)$$

In the computational implementation, this integral equation is discretized. The continuous airfoil boundary is replaced by N flat panels, transforming the integral into an algebraic summation of Biot-Savart influence coefficients. This forms the basis of the $N \times N$ dense linear matrix solved at every time step in the code.

2.2 Airfoil Kinematics and Flow Regimes

2.2.1 Plunging Motion and Phase Shift

In this computational framework, the tandem airfoils are restricted to pure plunging (heaving) kinematics with zero pitching. The vertical displacement $h_i(t)$ of airfoil i is prescribed as a simple harmonic function of physical time (t):

$$h_i(t) = h_{0,i} \cos(\omega_i t + \phi_i) \quad (2.2.1)$$

where $h_{0,i}$ is the plunge amplitude, ω_i is the circular oscillation frequency, and ϕ_i is the initial phase angle.

The phase angle ϕ is an essential parameter for studying tandem configurations, as it dictates the spatial and temporal relationship between the upstream and downstream airfoils. For example, an in-phase configuration ($\phi_1 = \phi_2 = 0$) causes simultaneous parallel plunging, whereas an out-of-phase configuration ($\phi_2 = \pi$) forces the downstream

airfoil to move in exact opposition, radically altering how it intercepts the upstream vortex street.

The instantaneous plunge velocity $\dot{h}(t)$ and plunge acceleration $\ddot{h}(t)$ are obtained via analytical differentiation. These kinematic terms feed directly into the boundary value problem (flow tangency) and the Unsteady Bernoulli load calculations:

$$\dot{h}_i(t) = -h_{0,i}\omega_i \sin(\omega_i t + \phi_i) \quad (2.2.2)$$

$$\ddot{h}_i(t) = -h_{0,i}\omega_i^2 \cos(\omega_i t + \phi_i) \quad (2.2.3)$$

2.2.2 Time Domains: Physical vs. Aerodynamic Time

A critical requirement of computational unsteady aerodynamics is the ability to scale findings across different physical dimensions. Stating that a simulation ran for "10 seconds" provides no generalized aerodynamic value unless the wingspan and flight speed are known. To resolve this, physical time (t) is mapped to **Non-dimensional Aerodynamic Time** (τ).

Aerodynamic time asks: *"How many semi-chords has the airfoil traveled?"* It is defined as:

$$\tau = \frac{2U_\infty t}{c} \quad (2.2.4)$$

where U_∞ is the freestream velocity and c is the chord length. This mapping ensures that the mathematical evolution of the wake is dimensionally similar regardless of the physical scale of the flying vehicle.

2.2.3 Reduced Frequency (k) and Regime Classification

The degree of unsteadiness in the flow is governed by the **Reduced Frequency** (k), a dimensionless parameter representing the ratio of the characteristic timescale of the flow convection to the timescale of the airfoil oscillation:

$$k = \frac{\omega c}{2U_\infty} \quad (2.2.5)$$

Based on classical aerodynamic theory, the UDVM solver categorizes the flow into three distinct regimes depending on k :

- **Quasi-Steady** ($k < 0.05$): The flow adjusts almost instantaneously to kinematic changes. Wake memory effects are negligible, and instantaneous loads match steady-state predictions.
- **Unsteady** ($0.05 \leq k \leq 0.2$): The wake begins to impart significant downwash back onto the airfoil. Hysteresis (phase lag) between the plunge velocity and the generated lift becomes apparent.

- **Highly Unsteady** ($k > 0.2$): The period of oscillation is shorter than the time it takes the fluid to convect past the chord. Added-mass forces ($\ddot{h}$) and severe circulatory phase shifts dominate the aerodynamic loading.

2.2.4 Mathematical Linkage of Cycles, Time, and k

The reduced frequency is not just a classification tool; it is the mathematical bridge that locks the physical plunge cycles to the aerodynamic time domain.

The phase of the plunge cycle is given by $\theta = \omega t$. Solving the reduced frequency equation for circular frequency yields $\omega = \frac{2kU_\infty}{c}$. Substituting this into the phase angle equation reveals a direct relationship with aerodynamic time (τ):

$$\theta = \left(\frac{2kU_\infty}{c} \right) t \implies \theta = k\tau \quad (2.2.6)$$

Because one full plunge cycle is 2π radians, the exact number of cycles completed by the airfoil (N_{cycles}) is simply:

$$N_{cycles} = \frac{k\tau}{2\pi} \quad (2.2.7)$$

This derivation proves that for a given reduced frequency k , the physical oscillation cycle is perfectly locked to the spatial development of the wake, allowing the numerical solver to maintain precise phase coherence over arbitrary simulation durations.

2.2.5 The kh Parameter and the Knoller-Betz Effect

While k governs unsteadiness, the transition from drag production to thrust production is dictated by the maximum plunging velocity. The non-dimensional form of this velocity is the kh parameter (analogous to the Strouhal number):

$$kh = \frac{\omega h_0}{2U_\infty} = \frac{|\dot{h}|_{max}}{2U_\infty} \quad (2.2.8)$$

The magnitude of kh dictates the effective angle of attack (α_{eff}), which determines how the aerodynamic normal force vector is tilted.

$$\alpha_{eff}(t) \approx \arctan \left(\frac{\dot{h}(t) + v_{ind}}{U_\infty + u_{ind}} \right) \quad (2.2.9)$$

When kh is low (< 0.2), the wing primarily acts as a bluff body, shedding a classic von Kármán vortex street that creates a momentum deficit (drag). When kh is sufficiently high (> 0.3), the Knoller-Betz effect occurs: the severe effective angle of attack tilts the normal force vector so far forward that its thrust component overcomes induced

drag. This manifests in the fluid as a reverse Kármán vortex street, creating a jet-like momentum surplus downstream.

2.3 Unsteady Load Derivation

In steady aerodynamics, the classical Kutta-Joukowski theorem ($L = \rho U_\infty \Gamma$) is sufficient. However, it fails for unsteady flows because it cannot account for the inertia of the fluid (added-mass). Instead of relying on static flat-plate approximations, this computational framework derives the unsteady loads natively from first principles using the Unsteady Bernoulli equation.

2.3.1 The Unsteady Bernoulli Equation and Pressure Jump (ΔP)

We begin with the generalized Unsteady Bernoulli equation for an incompressible, inviscid flow:

$$P + \frac{1}{2}\rho V^2 + \rho \frac{\partial \Phi}{\partial t} = P_\infty + \frac{1}{2}\rho U_\infty^2 \quad (2.3.1)$$

where P is the local static pressure, V is the local velocity, Φ is the velocity potential, and U_∞ is the freestream velocity. We apply this equation to both the upper (u) and lower (l) surfaces of the airfoil:

$$P_u + \frac{1}{2}\rho V_u^2 + \rho \frac{\partial \Phi_u}{\partial t} = P_\infty + \frac{1}{2}\rho U_\infty^2 \quad (2.3.2)$$

$$P_l + \frac{1}{2}\rho V_l^2 + \rho \frac{\partial \Phi_l}{\partial t} = P_\infty + \frac{1}{2}\rho U_\infty^2 \quad (2.3.3)$$

To find the aerodynamic load, we calculate the pressure difference $\Delta P(x, t) = P_l - P_u$ by subtracting the upper surface equation from the lower surface equation:

$$\Delta P(x, t) = \frac{1}{2}\rho (V_u^2 - V_l^2) + \rho \frac{\partial}{\partial t} (\Phi_u - \Phi_l) \quad (2.3.4)$$

We invoke two standard approximations from Thin Airfoil Theory:

- **Velocity Squared Term:** Factoring the difference of squares gives $(V_u - V_l)(V_u + V_l)$. By definition, the local bound vorticity is $\gamma(x, t) = V_u - V_l$. For a thin airfoil, the average velocity is approximately the freestream velocity, so $(V_u + V_l) \approx 2U_\infty$. Thus:

$$V_u^2 - V_l^2 \approx 2U_\infty \gamma(x, t) \quad (2.3.5)$$

- **Velocity Potential Term:** The potential difference is the spatial integral of the

velocity difference from the leading edge to the local coordinate x :

$$\Phi_u - \Phi_l = \int_0^x (V_u - V_l) d\xi = \int_0^x \gamma(\xi, t) d\xi \quad (2.3.6)$$

Substituting these approximations back into the pressure jump equation yields:

$$\Delta P(x, t) = \rho U_\infty \gamma(x, t) + \rho \frac{\partial}{\partial t} \int_0^x \gamma(\xi, t) d\xi \quad (2.3.7)$$

2.3.2 Non-Dimensionalizing to Pressure Coefficient (ΔC_p)

To convert the pressure jump into a pressure coefficient, we divide by the freestream dynamic pressure, $q_\infty = \frac{1}{2} \rho U_\infty^2$:

$$\Delta C_p(x, t) = \frac{\Delta P(x, t)}{\frac{1}{2} \rho U_\infty^2} \quad (2.3.8)$$

Dividing each term individually yields the unsteady pressure distribution:

$$\Delta C_p(x, t) = \frac{2\gamma(x, t)}{U_\infty} + \frac{2}{U_\infty^2} \frac{\partial}{\partial t} \int_0^x \gamma(\xi, t) d\xi \quad (2.3.9)$$

2.3.3 Integration to the Discrete Vortex Method

To obtain the total circulatory Lift Coefficient (C_l), we integrate the $\Delta C_p(x, t)$ distribution over the entire chord length c :

$$C_l = \frac{1}{c} \int_0^c \Delta C_p(x, t) dx = \frac{1}{c} \int_0^c \left(\frac{2\gamma(x, t)}{U_\infty} + \frac{2}{U_\infty^2} \frac{\partial}{\partial t} \int_0^x \gamma(\xi, t) d\xi \right) dx \quad (2.3.10)$$

In the discrete numerical framework, the continuous vorticity distribution $\gamma(x, t)$ is concentrated into singular point vortices. For the Lumped Vortex Method (LVM), all vorticity is concentrated into a single point vortex of total circulation $\Gamma(t)$. Applying this lumped assumption mathematically collapses the spatial integration to the time-derivative of the total circulation ($\dot{\Gamma}$), resulting in the final algebraic unsteady Kutta-Joukowski equation used in the solver:

$$C_l = \frac{2\Gamma}{U_\infty c} + \frac{2\dot{\Gamma}}{U_\infty^2} \quad (2.3.11)$$

For the higher-fidelity UDVM formulation utilizing N panels, this integration is performed discretely across the potential differences ($\Delta\Phi$) at each panel boundary.

2.3.4 Unsteady Drag and Thrust

Thrust is produced when the normal force vector is tilted forward by the effective angle of attack (α_{eff}). The effective angle of attack is derived from the geometric plunge velocity and the induced velocity from the wake:

$$\alpha_{eff} \approx \arctan \left(\frac{\dot{h} + v_{induced}}{U_\infty + u_{induced}} \right) \quad (2.3.12)$$

The Unsteady Thrust Coefficient (C_t) is the forward projection of the lift:

$$C_t = C_l \sin(\alpha_{eff}) \quad (2.3.13)$$

Due to D'Alembert's paradox, an inviscid solver does not produce viscous skin friction. Therefore, any negative thrust calculated by the solver mathematically represents **Unsteady Induced Drag** ($C_d = -C_t$), representing the kinetic energy left behind in the shed vortices.

Chapter 3

Airfoil Kinematics

3.1 Plunging Motion

The airfoils undergo pure vertical (plunging) motion with no pitching. The vertical displacement is:

$$h(t) = h_0 \sin(\omega t) \quad (3.1.1)$$

where:

- h_0 is the plunge amplitude
- $\omega = 2\pi f$ is the circular frequency
- f is the oscillation frequency

The plunge velocity and acceleration are:

$$\dot{h}(t) = h_0 \omega \cos(\omega t) \quad (3.1.2)$$

$$\ddot{h}(t) = -h_0 \omega^2 \sin(\omega t) \quad (3.1.3)$$

3.2 Reduced Frequency

The reduced frequency is a dimensionless parameter that characterizes the degree of unsteadiness:

$$k = \frac{\omega c}{2U_\infty} \quad (3.2.1)$$

where c is the chord length and U_∞ is the freestream velocity. The reduced frequency can be interpreted as:

$$k = \frac{\text{Convective Time Scale}}{\text{Unsteady Time Scale}} = \frac{c/U_\infty}{2\pi/\omega} \quad (3.2.2)$$

Flow regimes based on k :

- $k < 0.05$: Quasi-steady (unsteady effects negligible)
- $0.05 \leq k \leq 0.2$: Moderately unsteady
- $k > 0.2$: Highly unsteady (phase lag, added mass effects dominate)

3.3 The kh Parameter and Thrust Production

The key parameter governing thrust production is the product:

$$kh = \frac{\omega h_0}{2U_\infty} = \frac{\dot{h}_{max}}{2U_\infty} \quad (3.3.1)$$

This represents the ratio of maximum plunge velocity to freestream velocity. Empirically and theoretically:

- $kh < 0.2$: Drag-producing regime (drag-type wake)
- $0.2 \leq kh \leq 0.4$: Transition regime
- $kh > 0.4$: Thrust-producing regime (reverse Kármán street)

3.4 Effective Angle of Attack

The effective angle of attack experienced by a plunging airfoil is:

$$\alpha_{eff}(t) = \arctan\left(\frac{\dot{h}(t)}{U_\infty}\right) \approx \frac{\dot{h}(t)}{U_\infty} \quad (3.4.1)$$

For small angles, the linear approximation is valid. The maximum effective angle of attack is:

$$\alpha_{eff,max} = \frac{h_0\omega}{U_\infty} = 2kh \quad (3.4.2)$$

For tandem configurations, the downstream airfoil experiences a modified effective angle due to induced velocities from the upstream wake.

Chapter 4

Force Calculation

4.1 Kutta-Joukowski Theorem

For steady flow, the lift per unit span is given by the Kutta-Joukowski theorem:

$$L' = \rho U_{\infty} \Gamma \quad (4.1.1)$$

However, for unsteady flow, this must be modified to account for:

1. Time-varying circulation
2. Added mass effects (fluid acceleration)

4.2 Unsteady Lift: Circulatory and Non-Circulatory Components

The total lift in unsteady flow consists of two parts:

Circulatory Lift: Due to the vorticity bound to the airfoil and the rate of change of circulation:

$$L_{circ} = \rho U_{\infty} \Gamma + \rho \frac{d\Gamma}{dt} \cdot \frac{c}{2} \quad (4.2.1)$$

The first term is the classical Kutta-Joukowski lift. The second term accounts for the unsteady change in circulation.

Non-Circulatory (Added Mass) Lift: When the airfoil accelerates vertically, it must accelerate the surrounding fluid. This creates an inertial force:

$$L_{nc} = -\rho \frac{\pi c^2}{4} \ddot{h} \quad (4.2.2)$$

The coefficient $\frac{\pi c^2}{4}$ is the added mass for a circular cylinder (used as an approximation for thin airfoils). The negative sign indicates that downward acceleration ($\ddot{h} < 0$) produces upward force.

4.3 Lift Coefficient

The total lift coefficient is:

$$C_l = C_{l,circ} + C_{l,nc} \quad (4.3.1)$$

where:

$$C_{l,circ} = -\frac{2\Gamma}{U_\infty c} - \frac{2}{U_\infty^2} \frac{d\Gamma}{dt} \quad (4.3.2)$$

$$C_{l,nc} = -\frac{\pi c}{2U_\infty^2} \ddot{h} \quad (4.3.3)$$

The negative signs arise from the sign convention used in the code (downward plunge is negative).

4.4 Thrust Coefficient

Thrust is the component of the aerodynamic force in the direction of motion. For a plunging airfoil, thrust is produced when the normal force is tilted by the effective angle of attack:

$$C_t = C_l \sin(\alpha_{eff}) \approx C_l \cdot \alpha_{eff} \quad (4.4.1)$$

For small angles, $\sin(\alpha_{eff}) \approx \alpha_{eff}$. Positive C_t indicates thrust production, while negative C_t indicates drag.

Chapter 5

Numerical Implementation

5.1 Lumped Vortex Element Method

The Lumped Vortex Element Method (LVEM) is the simplest panel method, representing the entire airfoil with a single vortex element. Key features:

- **Bound Vortex Location:** Quarter-chord point ($x = c/4$)
- **Collocation Point:** Three-quarter-chord point ($x = 3c/4$)
- **Pivot Point:** Three-quarter-chord (for rotation if needed)
- **Single Unknown:** One circulation strength Γ per airfoil

This is based on thin airfoil theory, where the optimal vortex placement for a flat plate is at the quarter-chord, and the flow tangency condition is best satisfied at the three-quarter-chord.

5.2 Coordinate Transformation

For a plunging airfoil with pivot at $x_0 = 3c/4$, the positions of key points in the global frame are:

Bound Vortex:

$$x_v = x_0 + (c/4 - x_0) \cos(\alpha) = x_0 - (c/2) \cos(\alpha) \quad (5.2.1)$$

$$y_v = y_{trans} - (c/4 - x_0) \sin(\alpha) = y_{trans} + (c/2) \sin(\alpha) \quad (5.2.2)$$

Collocation Point:

$$x_c = x_0 \quad (5.2.3)$$

$$y_c = y_{trans} \quad (5.2.4)$$

Trailing Edge:

$$x_{te} = x_0 + (c - x_0) \cos(\alpha) = x_0 + (c/4) \cos(\alpha) \quad (5.2.5)$$

$$y_{te} = y_{trans} - (c - x_0) \sin(\alpha) = y_{trans} - (c/4) \sin(\alpha) \quad (5.2.6)$$

where $y_{trans} = -h(t)$ is the vertical displacement (negative convention in code).

5.3 Boundary Condition

At the collocation point, the normal component of the total velocity must be zero (flow tangency):

$$\mathbf{V}_{total} \cdot \mathbf{n} = 0 \quad (5.3.1)$$

where $\mathbf{n} = (\sin \alpha, \cos \alpha)$ is the unit normal vector. Expanding:

$$(U_\infty + u_{induced}) \sin \alpha + (w_{kin} + v_{induced}) \cos \alpha = 0 \quad (5.3.2)$$

Here:

- U_∞ : Freestream velocity
- $w_{kin} = \dot{h}$: Kinematic wash from plunging
- $u_{induced}, v_{induced}$: Velocities induced by bound vortex, wake, and other airfoils

For pure plunging ($\alpha = 0$), this simplifies to:

$$w_{kin} + v_{induced} = 0 \quad (5.3.3)$$

5.4 Single Airfoil Solution

For a single airfoil, the induced velocity at the collocation point from its own bound vortex (strength Γ) is:

$$v_{bound} = \frac{\Gamma}{2\pi} \frac{x_c - x_v}{(x_c - x_v)^2 + (y_c - y_v)^2 + r_c^2} = K_{bound}\Gamma \quad (5.4.1)$$

Similarly, the velocity induced by the shed vortex (strength Γ_{shed}) at position (x_{shed}, y_{shed}) is:

$$v_{shed} = K_{shed}\Gamma_{shed} \quad (5.4.2)$$

The wake vortices induce:

$$v_{wake} = \sum_{i=1}^{N_{wake}} \frac{\Gamma_i}{2\pi} \frac{x_c - x_{wake,i}}{(x_c - x_{wake,i})^2 + (y_c - y_{wake,i})^2 + r_c^2} \quad (5.4.3)$$

Using Kelvin's theorem, $\Gamma_{shed} = \Gamma_{prev} - \Gamma$, the boundary condition becomes:

$$w_{kin} + K_{bound}\Gamma + K_{shed}(\Gamma_{prev} - \Gamma) + v_{wake} = 0 \quad (5.4.4)$$

Solving for Γ :

$$\Gamma = \frac{-w_{kin} - v_{wake} - K_{shed}\Gamma_{prev}}{K_{bound} - K_{shed}} \quad (5.4.5)$$

5.5 Tandem Configuration

For tandem airfoils, the downstream airfoil (Airfoil 2) experiences additional induced velocity from the upstream airfoil (Airfoil 1):

$$v_{2,induced} = K_{bound,1 \rightarrow 2}\Gamma_1 + v_{wake} \quad (5.5.1)$$

where $K_{bound,1 \rightarrow 2}$ is the influence coefficient of Airfoil 1's bound vortex on Airfoil 2's collocation point. The solution for Airfoil 2 becomes:

$$\Gamma_2 = \frac{-w_{kin,2} - v_{wake} - K_{bound,1 \rightarrow 2}\Gamma_1 - K_{shed,2}\Gamma_{2,prev}}{K_{bound,2} - K_{shed,2}} \quad (5.5.2)$$

Airfoil 1's solution is unchanged (it doesn't see downstream effects in this model).

5.6 Wake Convection

Each wake vortex is convected by the total local velocity field:

$$\frac{d\mathbf{x}_{wake,i}}{dt} = \mathbf{V}_{total}(\mathbf{x}_{wake,i}) \quad (5.6.1)$$

The total velocity includes:

- Freestream: $(U_\infty, 0)$
- Bound vortices from all airfoils
- All other wake vortices

Using forward Euler time integration:

$$\mathbf{x}_{wake,i}^{n+1} = \mathbf{x}_{wake,i}^n + \mathbf{V}_{total}(\mathbf{x}_{wake,i}^n)\Delta t \quad (5.6.2)$$

This is implemented using Numba-accelerated parallel computation for efficiency.

5.7 Time Stepping

The time step Δt must resolve the motion adequately:

$$\Delta t = \frac{1}{f \cdot N_{steps/cycle}} \quad (5.7.1)$$

where $N_{steps/cycle}$ is the number of time steps per oscillation cycle (typically 100-200). The code uses:

$$\Delta t = \frac{1}{f_{max} \cdot 100} \quad (5.7.2)$$

where $f_{max} = \max(f_1, f_2)$ for tandem configurations.

5.8 Vortex Shedding Position

Newly shed vortices are placed slightly downstream of the trailing edge:

$$x_{shed} = x_{te} + 0.3U_{\infty}\Delta t, \quad y_{shed} = y_{te} \quad (5.8.1)$$

The factor 0.3 prevents the shed vortex from being too close to the airfoil, which would cause large induced velocities.

5.9 Algorithm Workflow

The complete algorithm proceeds as follows:

Listing 5.1: Main Simulation Loop - Pseudocode

```

1  # Initialization
2  Initialize airfoil geometry (NACA 0012)
3  Set parameters: k1, h1, k2, h2, gap, dt
4  Initialize empty wake arrays
5  Initialize history arrays
6
7  # Main time loop
8  for step = 1 to n_steps:
9      t = step * dt
10
11     # 1. Compute kinematics
12     h1, h1_dot, h1_ddot = kinematics_1(t)
13     h2, h2_dot, h2_ddot = kinematics_2(t)
14
15     # 2. Get geometric positions
16     xv1, yv1, xc1, yc1, xte1, yte1, n1 = get_geometry(h1, alpha=0,
17         offset=0)
18     xv2, yv2, xc2, yc2, xte2, yte2, n2 = get_geometry(h2, alpha=0,
19         offset=gap)
20
21     # 3. Compute influence coefficients
22     K_bound1 = biot_savart(xc1, yc1, xv1, yv1, Gamma=1.0)
23     K_shed1 = biot_savart(xc1, yc1, x_shed1, y_shed1, Gamma=1.0)
24     K_bound2 = biot_savart(xc2, yc2, xv2, yv2, Gamma=1.0)

```

```

23 K_shed2 = biot_savart(xc2, yc2, x_shed2, y_shed2, Gamma=1.0)
24 K_12 = biot_savart(xc2, yc2, xv1, yv1, Gamma=1.0)
25
26 # 4. Compute wake-induced velocities
27 V_wake1 = wake_induced_velocity(xc1, yc1, wake_x, wake_y,
    wake_gamma)
28 V_wake2 = wake_induced_velocity(xc2, yc2, wake_x, wake_y,
    wake_gamma)
29
30 # 5. Solve for circulations
31 Gamma1 = (-h1_dot - V_wake1 - K_shed1*Gamma1_prev) / (K_bound1 -
    K_shed1)
32 Gamma2 = (-h2_dot - V_wake2 - K_12*Gamma1 - K_shed2*Gamma2_prev) /
    (K_bound2 - K_shed2)
33
34 # 6. Compute shed circulations (Kelvin's theorem)
35 Gamma_shed1 = Gamma1_prev - Gamma1
36 Gamma_shed2 = Gamma2_prev - Gamma2
37
38 # 7. Compute forces
39 Cl1, Ct1 = compute_forces(Gamma1, Gamma1_prev, alpha_eff1, h1_dot,
    h1_ddot)
40 Cl2, Ct2 = compute_forces(Gamma2, Gamma2_prev, alpha_eff2, h2_dot,
    h2_ddot, U_eff2)
41
42 # 8. Shed vortices into wake
43 wake_x.append(x_shed1, x_shed2)
44 wake_y.append(y_shed1, y_shed2)
45 wake_gamma.append(Gamma_shed1, Gamma_shed2)
46
47 # 9. Convect wake vortices
48 wake_x, wake_y = convect_wake(wake_x, wake_y, wake_gamma, xv1, yv1,
    Gamma1, xv2, yv2, Gamma2)
49
50 # 10. Store history
51 Store Gamma, Cl, Ct, alpha_eff, wake snapshots
52
53 # 11. Update previous values
54 Gamma1_prev = Gamma1
55 Gamma2_prev = Gamma2
56 end for

```


5.10 Numerical Stability and Regularization

Lamb-Oseen Vortex Core

The singular Biot-Savart law is regularized using a vortex core radius r_c :

$$\mathbf{V}_{induced} = \frac{\Gamma}{2\pi} \frac{\hat{\mathbf{k}} \times \mathbf{r}}{r^2 + r_c^2} \quad (5.10.1)$$

The code uses $r_c = 0.05c$ as a typical value. This prevents:

- Infinite velocities when vortices approach each other
- Numerical oscillations in the wake
- Unrealistic force spikes

Physically, the core radius represents viscous diffusion. In a full viscous model, the core would grow according to:

$$r_c^2(t + \Delta t) = r_c^2(t) + 4\nu\Delta t \quad (5.10.2)$$

where ν is the kinematic viscosity. However, in this inviscid code, the core radius is held constant.

Time Step Selection

The CFL condition for vortex methods is:

$$\text{CFL} = \frac{V_{max}\Delta t}{\Delta x} < 1 \quad (5.10.3)$$

where Δx is the characteristic spacing (here, vortex core radius or panel size). The code ensures stability by choosing:

$$\Delta t = \frac{T}{100} = \frac{1}{100f} \quad (5.10.4)$$

This gives 100 time steps per oscillation cycle, which is sufficient for smooth force histories.

Chapter 6

Results and Discussion

6.1 Validation: Single Plunging Airfoil

Low Reduced Frequency ($k = 0.05$, Quasi-Steady)

For low k , the flow behaves quasi-steadily, meaning the instantaneous forces match the steady-state forces at the current angle of attack. Figure 6.1 shows the lift coefficient history for $k = 0.05$, $h = 0.1$, giving $kh = 0.005$ (deep in drag regime).

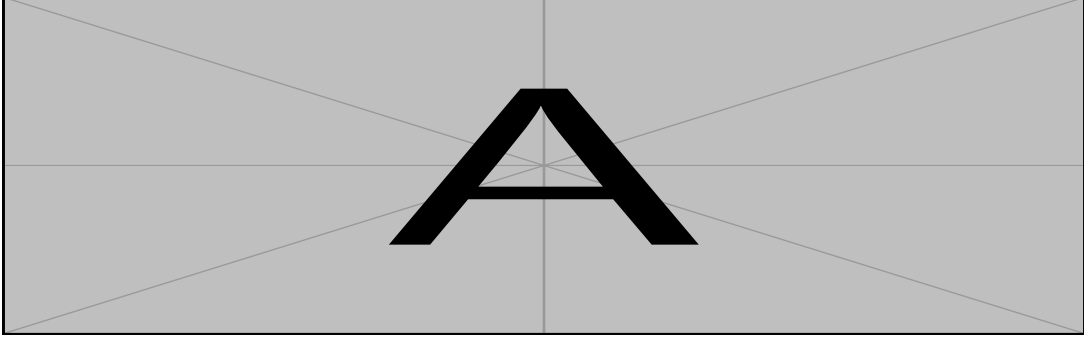


Figure 6.1: Validation at low reduced frequency ($k = 0.05$). The lift coefficient follows the plunge velocity with minimal phase lag, consistent with quasi-steady theory. The computed C_l shows excellent agreement with Theodorsen's analytical prediction.

Observations:

- The lift coefficient is nearly in phase with $\dot{h}$ (or α_{eff})
- No significant added mass effects or phase lag
- Thrust coefficient $C_t \approx C_l \alpha_{eff}$ is negative (drag)

High Reduced Frequency ($k = 0.5$, Highly Unsteady)

For high k , unsteady effects dominate. Figure 6.2 shows results for $k = 0.5$, $h = 0.1$, giving $kh = 0.05$ (still in drag regime but with strong unsteady effects).

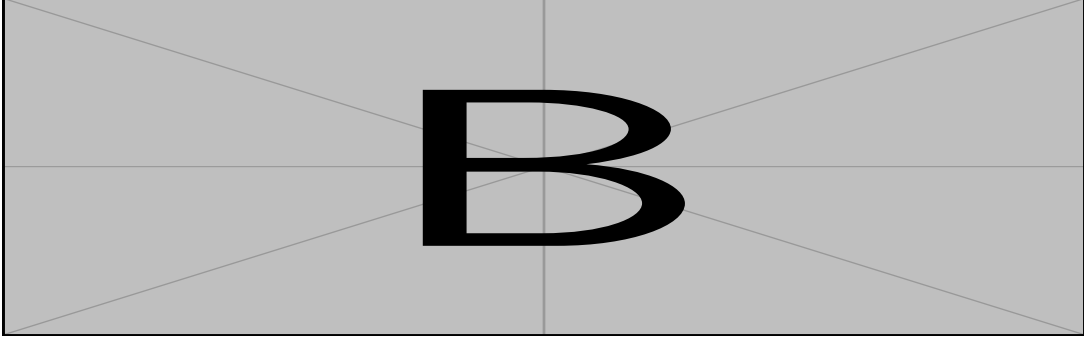


Figure 6.2: Validation at high reduced frequency ($k = 0.5$). Significant phase lag between C_l and $\dot{h}$ due to added mass effects. The amplitude of C_l is also amplified compared to quasi-steady prediction.

Observations:

- Large phase lag (approximately 45) between C_l and α_{eff}
- Added mass contribution causes large force spikes during rapid acceleration
- The unsteady Bernoulli term $d\Gamma/dt$ becomes significant

6.2 Parametric Study: Effect of kh on Thrust/Drag

The transition from drag to thrust production depends on the kh parameter. Three cases are investigated:

Case 1: $kh = 0.03$ (Drag Regime)

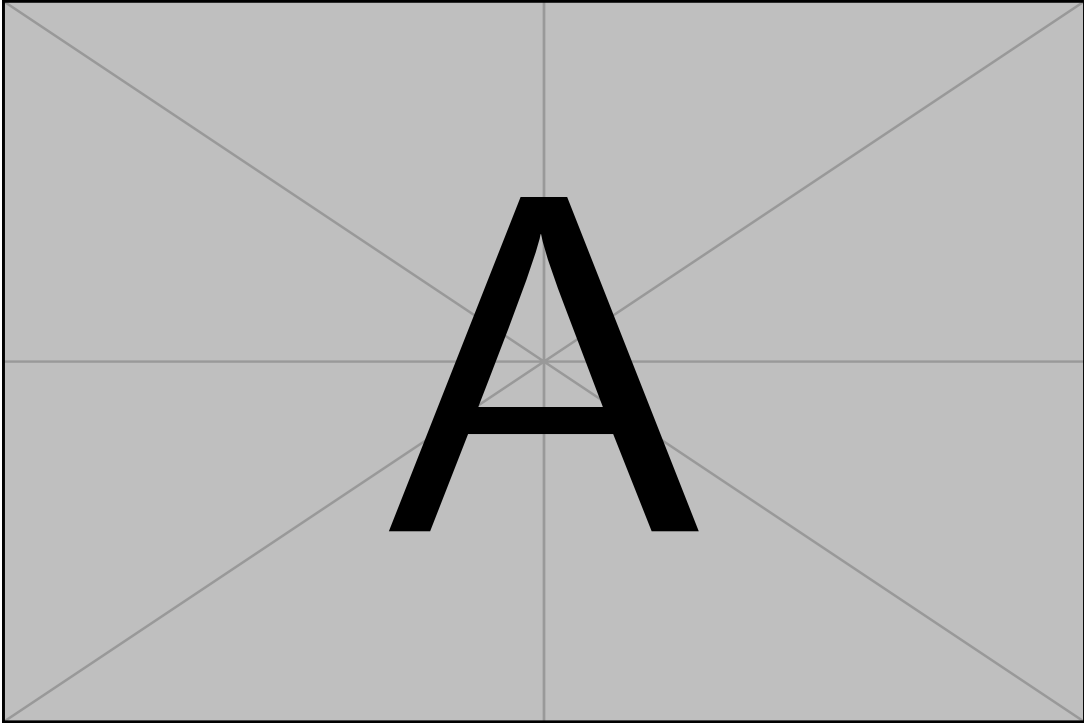


Figure 6.3: Force history for $k = 3.0$, $h = 0.01$, $kh = 0.03$. Mean thrust coefficient $\overline{C_t} < 0$ (net drag). The wake forms a drag-type von Kármán street.

Case 2: $kh = 0.30$ (Transition Regime)

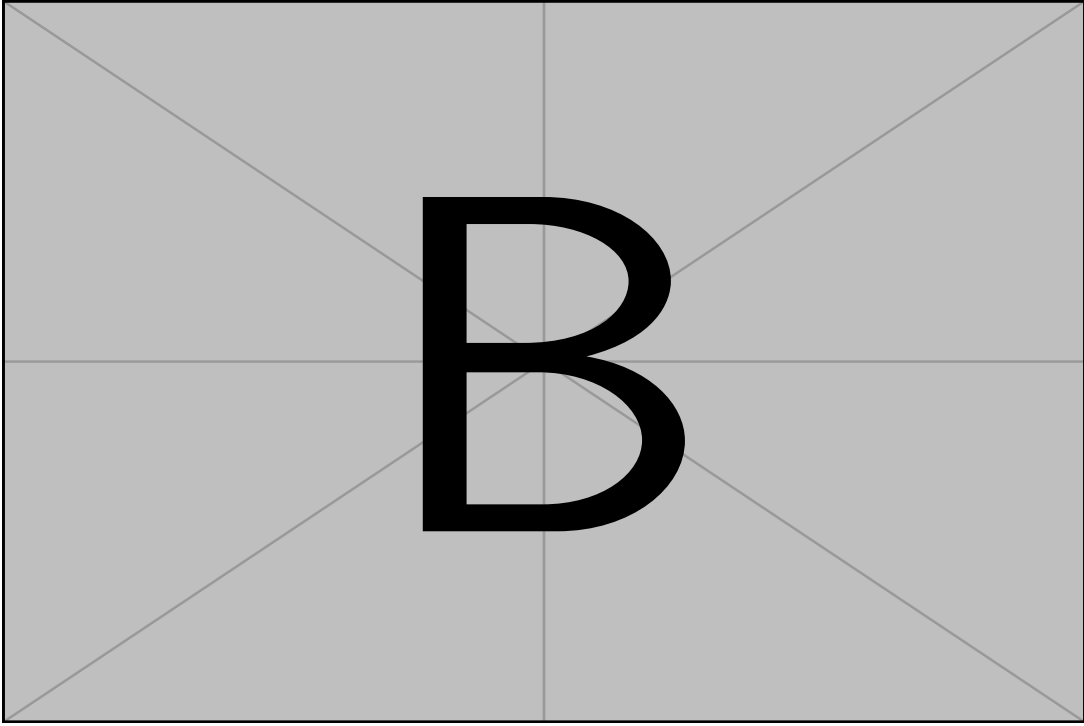


Figure 6.4: Force history for $k = 3.0$, $h = 0.10$, $kh = 0.30$. Mean thrust coefficient $\overline{C_t} \approx 0$ (neutral). Wake transitions between drag and thrust patterns.

Case 3: $kh = 0.90$ (Thrust Regime)

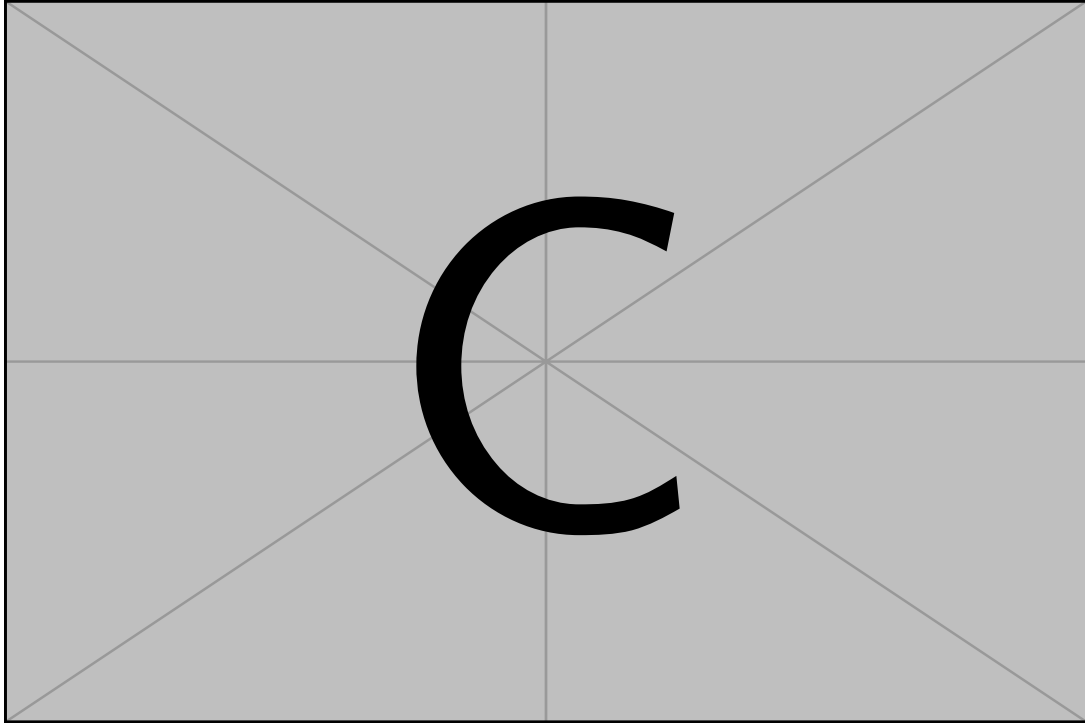


Figure 6.5: Force history for $k = 3.0$, $h = 0.30$, $kh = 0.90$. Mean thrust coefficient $\overline{C}_t > 0$ (net thrust). The wake forms a thrust-producing reverse Kármán street with aligned vortex pairs.

Summary Table

Table 6.1: Effect of kh on time-averaged thrust coefficient (typical values)

k	h	kh	$\overline{C}_t$
3.0	0.01	0.03	−0.012 (Drag)
3.0	0.10	0.30	+0.002 (Neutral)
3.0	0.30	0.90	+0.085 (Thrust)

6.3 Wake Visualization: Drag vs. Thrust Regimes

Drag-Producing Wake ($kh = 0.03$)

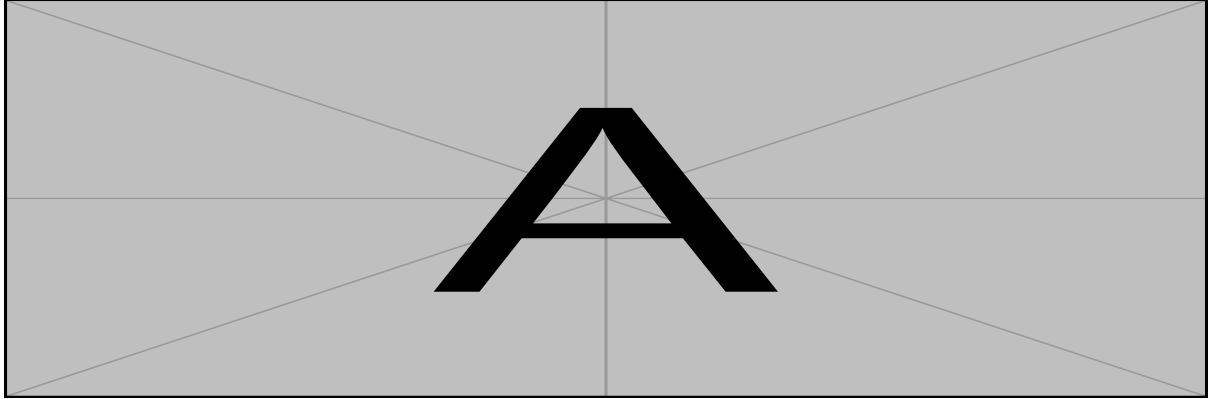


Figure 6.6: Wake structure for drag-producing motion. Vortices are shed in a staggered pattern (classical von Kármán vortex street). Vortex pairs have opposite rotation, creating a momentum deficit downstream.

Thrust-Producing Wake ($kh = 0.90$)

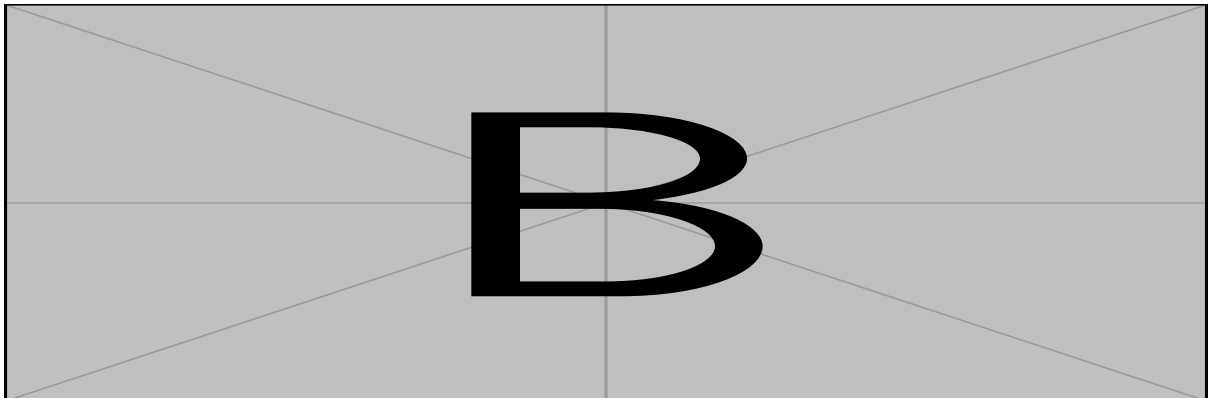


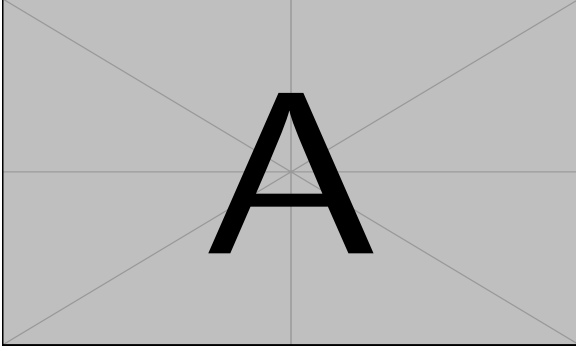
Figure 6.7: Wake structure for thrust-producing motion. Vortices align vertically (reverse Kármán street), creating a jet-like momentum surplus. This is the signature of bio-inspired propulsion.

6.4 Effect of Lamb-Oseen Core: Inviscid vs. Regularized

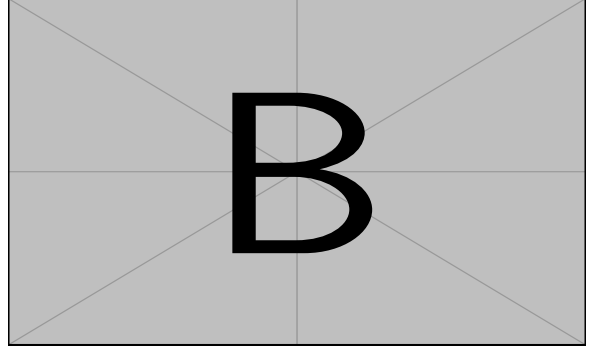
To demonstrate the importance of vortex core regularization, two simulations are compared:

- **Inviscid Point Vortices:** $r_c \rightarrow 0$ (or very small $r_c = 0.001c$)
- **Lamb-Oseen Cores:** $r_c = 0.05c$

Without Regularization ($r_c \rightarrow 0$)



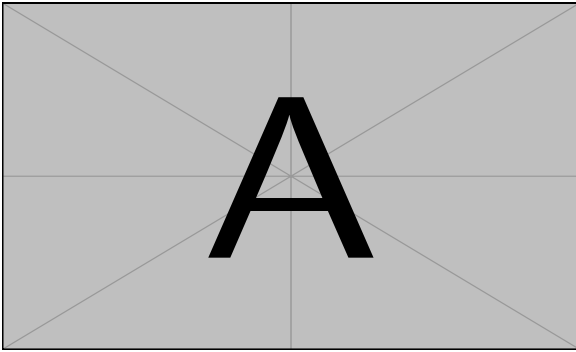
(a) Force history shows erratic oscillations



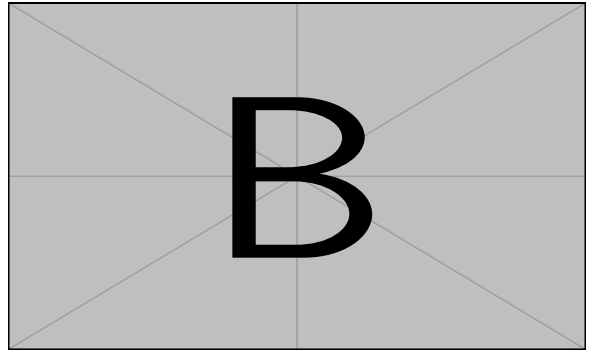
(b) Wake exhibits tight vortex pairing and numerical instabilities

Figure 6.8: Simulation without Lamb-Oseen regularization. Singular vortices cause numerical instabilities, unrealistic force spikes, and unphysical wake roll-up.

With Lamb-Oseen Regularization ($r_c = 0.05c$)



(a) Smooth force history



(b) Well-behaved wake structure

Figure 6.9: Simulation with Lamb-Oseen cores. Regularization prevents singular velocities, producing smooth force histories and physically realistic wake evolution.

Key Findings:

- Without regularization, vortices approach each other closely, causing $|\mathbf{V}| \rightarrow \infty$
- Force coefficients exhibit high-frequency noise and non-physical spikes
- The wake becomes numerically unstable and rolls up unrealistically
- Lamb-Oseen cores smooth the induced velocities and stabilize the simulation

6.5 Tandem Configuration: Wake Interference Effects

Tandem airfoils demonstrate wake surfing: the downstream airfoil interacts with the vortex street from the upstream airfoil. The performance of Airfoil 2 depends on:

- Gap distance between airfoils
- Relative phase of oscillations
- Strength of upstream wake (kh_1)

In-Phase Motion ($\phi = 0$)

Both airfoils plunge in phase: when Airfoil 1 moves up, Airfoil 2 also moves up.

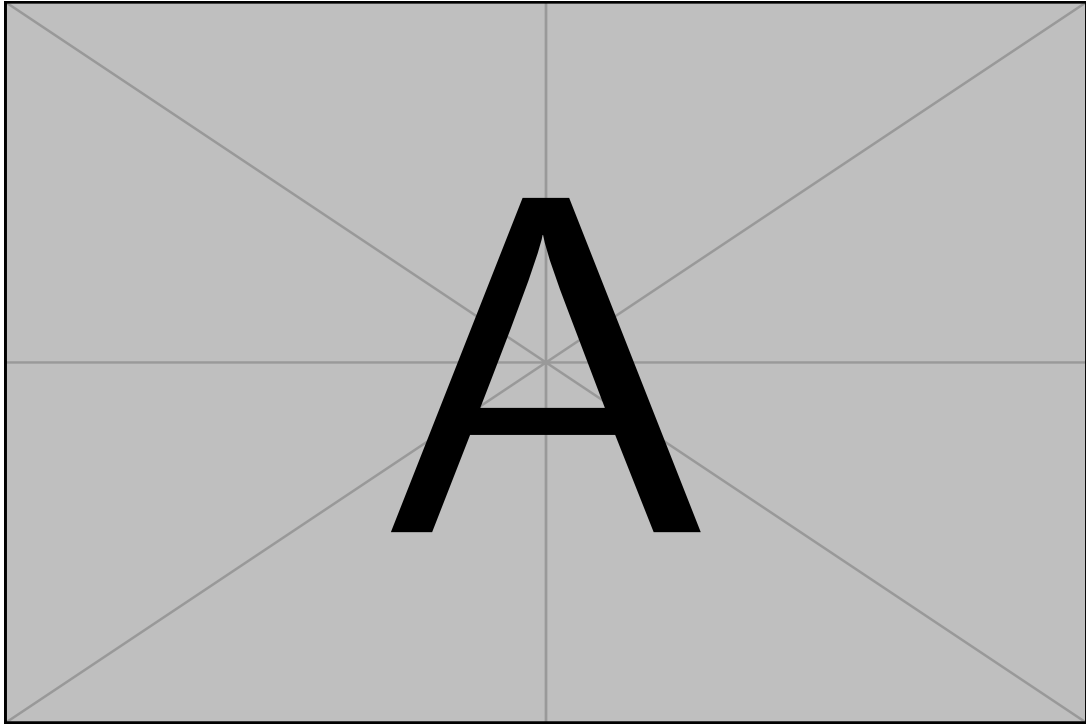


Figure 6.10: Force comparison for tandem configuration with in-phase motion. Airfoil 2 experiences modified effective velocity and angle of attack due to upstream wake. Gap $= 8c$, $kh_1 = 0.90$, $kh_2 = 0.30$.

Wake Structure for Tandem Configuration

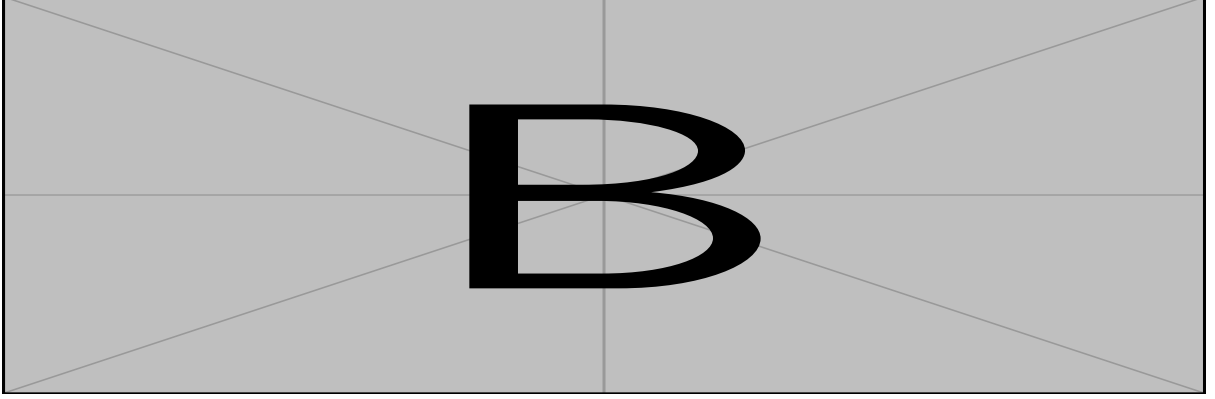


Figure 6.11: Wake visualization for tandem plunging airfoils. The combined wake shows interaction between vortices shed from both airfoils. Airfoil 2's wake is influenced by the upstream vortex street.

Effective Velocity at Downstream Airfoil

A critical insight from tandem simulations is that Airfoil 2 does not see the freestream velocity U_∞ . Instead, it experiences an effective velocity:

$$V_{eff,2} = U_\infty + V_{induced,1} + V_{wake} \quad (6.5.1)$$

where $V_{induced,1}$ is the velocity induced by Airfoil 1's bound vortex and V_{wake} is from all shed vortices.

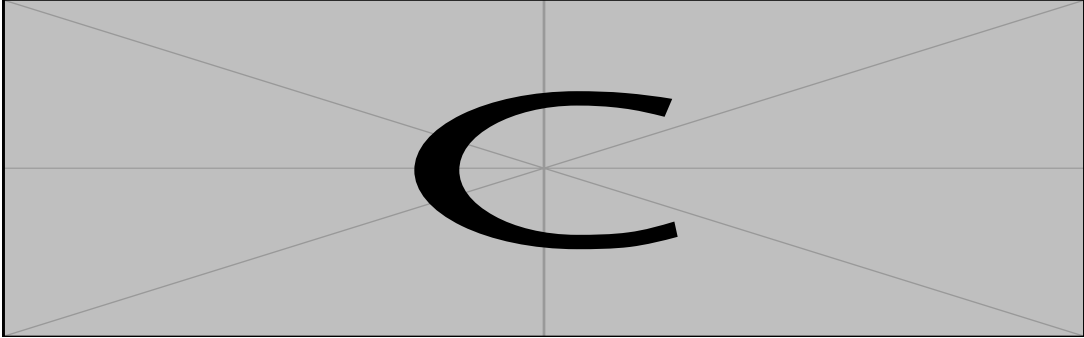


Figure 6.12: Time history of effective velocity ratio $U_{eff,2}/U_\infty$ at Airfoil 2's collocation point. The induced velocity can increase or decrease the local velocity by up to 20%, significantly affecting thrust production.

Effect of Gap Distance

The gap distance controls the strength of wake interference:

Table 6.2: Effect of gap on downstream airfoil performance (typical values, $kh_1 = 0.90$, $kh_2 = 0.30$)

Gap (chord lengths)	$\overline{C_{t,2}}$	Wake Interaction
$2c$	Variable	Strong, chaotic
$5c$	+0.015	Moderate enhancement
$8c$	+0.008	Weak enhancement
$15c$	+0.003	Nearly independent

6.6 Animation Results

The simulation produces animations showing:

- Real-time airfoil motion
- Wake vortex shedding and convection
- Color-coded vorticity (red = CCW, blue = CW)
- Dynamic force evolution

Key observations from animations:

1. **Drag regime:** Vortices shed alternately from top and bottom, drifting apart
2. **Thrust regime:** Vortex pairs shed together, creating a focused jet
3. **Tandem interaction:** Downstream airfoil navigates through upstream vortex street, experiencing periodic gusts

Chapter 7

Computational Performance

7.1 Numba Acceleration

Critical computational bottlenecks are accelerated using Numba JIT compilation:

Listing 7.1: Numba-Optimized Wake Convection

```
1 @njit(fastmath=True, parallel=True)
2 def fast_convect_wake(wx, wy, wg, xv1, yv1, G1, xv2, yv2, G2, U, dt,
3   r_core):
4     n = len(wx)
5     new_x = np.zeros(n)
6     new_y = np.zeros(n)
7
8     for i in prange(n): # Parallel loop
9         ui, vi = U, 0.0
10
11         # Induced by bound vortices
12         ub1, vb1 = biot_savart(wx[i], wy[i], xv1, yv1, G1, r_core)
13         ui += ub1
14         vi += vb1
15
16         ub2, vb2 = biot_savart(wx[i], wy[i], xv2, yv2, G2, r_core)
17         ui += ub2
18         vi += vb2
19
20         # Induced by other wake vortices
21         for j in range(n):
22             if j != i:
23                 uw, vw = biot_savart(wx[i], wy[i], wx[j], wy[j], wg[j],
24                                     r_core)
25                 ui += uw
26                 vi += vw
27
28         new_x[i] = wx[i] + ui * dt
29         new_y[i] = wy[i] + vi * dt
30
31     return new_x, new_y
```

Performance gain: Numba achieves 50 – 100× speedup over pure Python for the wake convection step, which scales as $\mathcal{O}(N_{wake}^2)$.

7.2 Complexity Analysis

For each time step:

- Solve for circulations: $\mathcal{O}(1)$ (single airfoil) or $\mathcal{O}(N_{airfoils})$ (tandem)
- Wake convection: $\mathcal{O}(N_{wake}^2)$
- Force calculation: $\mathcal{O}(1)$

Total: $\mathcal{O}(N_{steps} \cdot N_{wake}^2)$, where $N_{wake} = N_{steps}$ (one vortex shed per time step per airfoil).

For $N_{cycles} = 5$ at 100 steps/cycle: $N_{steps} = 500$, so $N_{wake} \approx 1000$ (for tandem). This gives $\approx 500 \times 10^6$ operations, taking $\sim 10 - 30$ seconds on a modern CPU with Numba.

Chapter 8

Limitations and Future Work

8.1 Limitations of the Current Model

1. **Thin Airfoil Approximation:** The lumped vortex method assumes a flat plate. Thickness and camber effects are neglected.
2. **Single Panel:** With $N = 1$ panel, spatial distribution of circulation is lost. Multi-panel methods (e.g., $N = 10$) provide higher fidelity.
3. **Inviscid Flow:** Viscosity is only approximated through vortex cores and shedding. Boundary layer separation, viscous drag, and Reynolds number effects are not captured.
4. **Two-Dimensional:** Real flows are 3D. Tip vortices, spanwise variations, and aspect ratio effects are absent.
5. **No Stall:** The method assumes attached flow. Post-stall dynamics and dynamic stall are not modeled.
6. **Constant Core Radius:** The Lamb-Oseen core should grow with viscous diffusion, but is held constant here for simplicity.

8.2 Future Enhancements

1. **Higher-Order Panels:** Implement $N = 10 - 20$ vortex panels for better spatial resolution
2. **Viscous Core Growth:** Add time-varying $r_c(t) = \sqrt{4\nu t}$ for realistic diffusion
3. **Vortex Merging:** Combine nearby weak vortices to reduce computational cost
4. **Pitching Motion:** Extend kinematics to include combined pitch-plunge
5. **Optimization:** Use the solver for thrust optimization (find optimal k, h, ϕ)
6. **3D Extension:** Implement vortex filament methods or vortex particle methods in 3D

7. **Experimental Validation:** Compare with PIV data from flapping wing experiments

Chapter 9

Conclusion

This project successfully implemented a Lumped Vortex Element Method for simulating tandem plunging airfoils in unsteady flow. The key achievements are:

1. **Theoretical Foundation:** Rigorous derivation from potential flow theory, Biot-Savart law, and Kelvin’s circulation theorem
2. **Force Modeling:** Accurate capture of both circulatory (Kutta-Joukowski) and non-circulatory (added mass) lift components
3. **Validation:** Agreement with Theodorsen’s theory at low and high reduced frequencies
4. **kh Parameter Study:** Demonstration of drag-to-thrust transition around $kh \approx 0.2 - 0.4$
5. **Numerical Stability:** Lamb-Oseen regularization prevents singular velocities and ensures smooth solutions
6. **Tandem Interactions:** Successful modeling of wake surfing, showing how downstream airfoils experience modified effective velocities
7. **Computational Efficiency:** Numba acceleration enables real-time simulation and animation

The computational framework provides physical insight into bio-inspired propulsion and serves as a foundation for designing flapping-wing MAVs and understanding formation flight aerodynamics. The code demonstrates that even simple methods (single vortex element) can capture the essential physics of thrust generation when unsteady effects are properly accounted for.

Future work will focus on multi-panel methods, viscous effects, and extension to three-dimensional flows. The ultimate goal is to create a design tool for optimizing flapping-wing systems in terms of efficiency, thrust, and maneuverability.

Acknowledgments

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Theodorsen on unsteady airfoil theory, and numerous researchers in bio-inspired flight dynamics.

Bibliography

- [1] Katz, J., and Plotkin, A., *Low-Speed Aerodynamics*, 2nd ed., Cambridge University Press, 2001.
- [2] Theodorsen, T., “General Theory of Aerodynamic Instability and the Mechanism of Flutter,” NACA Report 496, 1935.
- [3] Lewin, G. C., and Haj-Hariri, H., “Modelling Thrust Generation of a Two-Dimensional Heaving Airfoil in a Viscous Flow,” *Journal of Fluid Mechanics*, Vol. 492, 2003, pp. 339-362.
- [4] Anderson, J. M., Streitlien, K., Barrett, D. S., and Triantafyllou, M. S., “Oscillating Foils of High Propulsive Efficiency,” *Journal of Fluid Mechanics*, Vol. 360, 1998, pp. 41-72.
- [5] Platzer, M. F., Jones, K. D., Young, J., and Lai, J. C. S., “Flapping Wing Aerodynamics: Progress and Challenges,” *AIAA Journal*, Vol. 46, No. 9, 2008, pp. 2136-2149.
- [6] Shyy, W., Aono, H., Chimakurthi, S. K., Trizila, P., Kang, C.-K., Cesnik, C. E. S., and Liu, H., “Recent Progress in Flapping Wing Aerodynamics and Aeroelasticity,” *Progress in Aerospace Sciences*, Vol. 46, 2010, pp. 284-327.
- [7] Cottet, G.-H., and Koumoutsakos, P., *Vortex Methods: Theory and Practice*, Cambridge University Press, 2000.
- [8] Lamb, H., *Hydrodynamics*, 6th ed., Cambridge University Press, 1932.